Lab 21 — K-Nearest Neighbour Classifier (Iris Dataset)

Aim

To implement the K-Nearest Neighbour (KNN) algorithm to classify the Iris dataset and evaluate the prediction performance.

Algorithm / Procedure

1. Load the Iris dataset.
2. Split the dataset into training and testing sets.
3. Create a KNN classifier with k = 5.
4. Train the classifier using the training data.
5. Predict the classes of the test data.
6. Calculate the accuracy.
7. Display the classification report.

Program

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score,classification_report

iris=load_iris()

X_train,X_test,y_train,y_test=train_test_split(iris.data,iris.target,test_size=0.3,random_state=
1)
```

```
model=KNeighborsClassifier(n_neighbors=5)

model.fit(X_train,y_train)

y_pred=model.predict(X_test)

print("Accuracy:",accuracy_score(y_test,y_pred))

print(classification_report(y_test,y_pred,target_names=iris.target_names))
```

Output

```
Accuracy: 0.9777777777777777
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        14
  versicolor       0.95      1.00      0.97        18
   virginica       1.00      0.92      0.96        13

    accuracy                           0.98        45
   macro avg       0.98      0.97      0.98        45
weighted avg       0.98      0.98      0.98        45
```

Result

The K-Nearest Neighbour classifier was successfully implemented using k = 5 and achieved approximately 97.8% accuracy on the Iris dataset.